Test Plan: Customization Tools (Brands, Themes, Custom Pages, Custom Domains, Custom Email Templates, Email Domains)

1. Executive Summary

This document outlines a comprehensive test plan for validating prompt inputs and parameter handling across the Okta MCP Server's customization tools. The test plan is focused on prompt validation scenarios (positive, negative, edge cases, and scale tests) without requiring integration with live Okta APIs, unless absolutely critical to verify core functionality.

The customization tools covered in this plan are:

- Brands
- Themes
- Custom Pages
- Custom Domains
- Custom Email Templates
- Email Domains

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2. Test Strategy and Scope

2.1 Testing Approach

This test plan employs a multi-layered validation strategy:

1. **Prompt Validation Tests:** Verify that user prompts are correctly parsed and validated before API calls
2. **Parameter Format Tests:** Ensure input parameters conform to expected formats and constraints
3. **Naming Conflict Tests:** Confirm duplicate detection and appropriate error messages
4. **Error Handling Tests:** Validate correct error messages for various failure scenarios
5. **Acceptance Criteria Validation:** Verify compliance with specified acceptance criteria

2.2 In-Scope Test Categories

- Input parameter validation (format, length, special characters)
- Brand name and template name uniqueness checks
- Sensitive information handling (no exposure of secret keys/tokens)
- ID validation and path traversal prevention
- Status value validation against enum constraints
- Email format validation
- Domain name validation
- Certificate source type validation
- Language tag validation (IETF BCP 47)
- Pagination parameter validation
- Expansion parameter validation

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3. Test Categories

3.1 Positive Scenarios

These tests verify that valid inputs produce expected results or appropriate behavior.

3.1.1 Brands - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
BRAND-POS-001	List all brands	No parameters	Returns list of brands with pagination metadata	Basic retrieval
BRAND-POS-002	List brands with expand	expand=["themes", "emailDomain"]	Returns brands with nested theme and email domain data	Multiple expand values
BRAND-POS-003	Search brand by name	q="Default"	Returns brands matching search term	Case-insensitive search

BRAND-POS-004	List with limit	limit=50	Returns maximum 50 brands	Respects limit constraint
BRAND-POS-005	Pagination with cursor	after="cursor123"	Returns next page of results	Valid cursor format
BRAND-POS-006	Fetch all brands	fetch_all=True	Returns all brands across all pages	Automatic pagination
BRAND-POS-007	Get specific brand	brand_id="bnd114iNkrcN6aR680g4"	Returns single brand object	Valid brand ID format
BRAND-POS-008	Create brand	brand_id auto-generated, name="NewBrand"	Brand created successfully	New unique brand
BRAND-POS-009	Replace brand	brand_id="bnd123", name="UpdatedBrand"	Brand updated successfully	Existing brand modified
BRAND-POS-010	List brand domains	brand_id="bnd123"	Returns domains for brand	Brand-specific query
BRAND-POS-011	Delete brand	brand_id="bnd123" with user confirmation	Brand deleted successfully	Destructive operation

3.1.2 Themes - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
THEME-POS-001	List themes	brand_id="bnd123"	Returns single theme object	One theme per brand

THEME-POS-002	Get specific theme	brand_id="bnd123", theme_id="thdul904tTZ6kWVhP0g3"	Returns theme with full configuration	Valid theme ID
THEME-POS-003	Update theme colors	brand_id="bnd123", theme_id="thd123", primaryColor="#FF0000"	Theme updated	Color format accepted
THEME-POS-004	Update multiple theme properties	Provides colors, logo, CSS, variants	Theme updated with all changes	Complex update
THEME-POS-005	Upload logo	brand_id="bnd123", theme_id="thd123", file_path="logo.png"	Logo uploaded (< 100 kB)	Valid image format
THEME-POS-006	Upload favicon	brand_id="bnd123", theme_id="thd123", file_path="favicon.ico"	Favicon uploaded (< 2 MB)	Valid icon format
THEME-POS-007	Upload background image	brand_id="bnd123", theme_id="thd123", file_path="bg.jpg"	Background uploaded (< 2 MB)	Valid image format
THEME-POS-008	Set sign-in touchpoint variant	variant="BACKGROUND_IMAGE"	Theme variant set	Valid enum value
THEME-POS-009	Set dashboard variant	variant="FULL_THEME"	Variant set for dashboard	Valid enum value
THEME-POS-010	Set email template variant	variant="FULL_THEME"	Variant set for email	Valid enum value
THEME-POS-011	Delete logo with confirmation	brand_id="bnd123", theme_id="thd123"	Logo deleted, reverts to default	Destructive with confirmation

THEME-POS-012	Delete favicon with confirmation	brand_id="bnd123", theme_id="thd123"	Favicon deleted, reverts to default	Destructive with confirmation
THEME-POS-013	Delete background image	brand_id="bnd123", theme_id="thd123"	Background deleted	Destructive with confirmation

3.1.3 Custom Pages - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
PAGE-POS-001	Get error page resources	brand_id="bnd123"	Returns _links to error page variants	Navigation structure
PAGE-POS-002	Get customized error page	brand_id="bnd123"	Returns custom error page HTML	Existing customization
PAGE-POS-003	Get default error page	brand_id="bnd123"	Returns Okta default error page	Read-only reference
PAGE-POS-004	Get preview error page	brand_id="bnd123"	Returns preview error page	Testing environment
PAGE-POS-005	Update customized error page	brand_id="bnd123", page_content="..."	Error page updated	New HTML accepted
PAGE-POS-006	Update preview error page	brand_id="bnd123", page_content="..."	Preview page updated	Sandbox update

PAGE-POS-007	Get error page resources expanded	brand_id="bnd123", expand=["customized", "default"]	Returns embedded page content	Expanded resources
PAGE-POS-008	Get sign-in page resources	brand_id="bnd123"	Returns sign-in page variants	Sign-in specific
PAGE-POS-009	Get customized sign-in page	brand_id="bnd123"	Returns custom sign-in HTML	Existing customization
PAGE-POS-010	Update customized sign-in page	brand_id="bnd123", page_content="..."	Sign-in page updated	New HTML accepted
PAGE-POS-011	Update widget customizations	brand_id="bnd123", widget_customizations={...}	Widget settings applied	Widget config accepted
PAGE-POS-012	Set content security policy	brand_id="bnd123", csp_mode="enforced"	CSP policy set	Policy mode valid
PAGE-POS-013	List sign-in widget versions	brand_id="bnd123"	Returns available widget versions	Version listing
PAGE-POS-014	Get sign-out page settings	brand_id="bnd123"	Returns sign-out configuration	Current settings
PAGE-POS-015	Update sign-out page type	brand_id="bnd123", type="EXTERNALLY_HOSTED", url="https://..."	Sign-out config updated	External redirect
PAGE-POS-016	Delete customized error page	brand_id="bnd123" with confirmation	Page deleted, reverts to default	Destructive operation

PAGE-POS-017	Delete preview error page	brand_id="bnd123" with confirmation	Preview deleted	Sandbox reset
PAGE-POS-018	Delete customized sign-in page	brand_id="bnd123" with confirmation	Page deleted, reverts to default	Destructive operation

3.1.4 Custom Domains - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
DOMAIN-POS-001	List custom domains	No parameters	Returns all custom domains including "default"	All domains returned
DOMAIN-POS-002	Get specific domain	domain_id="dom123"	Returns single domain with status	Valid domain ID
DOMAIN-POS-003	Create domain OKTA_MANAGED	domain="login.example.com", type="OKTA_MANAGED"	Domain created with DNS records	Type validated
DOMAIN-POS-004	Create domain MANUAL	domain="auth.example.com", type="MANUAL"	Domain created, manual cert required	Type validated
DOMAIN-POS-005	Domain validation status	domain_id="dom123"	Returns validationStatus	Status enum respected
DOMAIN-POS-006	Domain DNS records present	domain_id="dom123"	Returns dnsRecords with TXT and CNAME	DNS data structure

DOMAIN-POS-007	Verify domain	domain_id="dom123"	Verification initiated	DNS records must be configured first
DOMAIN-POS-008	Update domain	domain_id="dom123", brandId="bnd123"	Domain association updated	Brand link modified
DOMAIN-POS-009	Upload certificate	domain_id="dom123", cert_type="PEM", source="MANUAL"	Certificate uploaded	MANUAL source type
DOMAIN-POS-010	Delete domain	domain_id="dom123" with confirmation	Domain deleted	Destructive operation

3.1.5 Custom Email Templates - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
TEMPLATE-POS-001	List email templates	brand_id="bnd123"	Returns all template names	Template enumeration
TEMPLATE-POS-002	Get email template	brand_id="bnd123", template_name="UserActivation"	Returns template with links	Valid template name
TEMPLATE-POS-003	List customizations	brand_id="bnd123", template_name="ForgotPassword"	Returns all language variants	Multiple languages
TEMPLATE-POS-004	Get customization	brand_id="bnd123", template_name="PasswordResetByAdmin", customization_id="id123"	Returns single variant	Specific customization
TEMPLATE-POS-005	Create customization	brand_id="bnd123", template_name="AccountLockout",	Customization created	French version created

		language="fr", subject="...", body="..."		
TEMPLATE-POS-006	Replace customization	customization_id="id123", subject="new subject"	Customization updated	Existing variant updated
TEMPLATE-POS-007	Set default customization	customization_id="id123"	Customization marked as default	isDefault flag set
TEMPLATE-POS-008	Get default content	brand_id="bnd123", template_name="UserActivation"	Returns Okta default template	Reference content
TEMPLATE-POS-009	Get customization preview	customization_id="id123"	Returns rendered template preview	Preview generation
TEMPLATE-POS-010	Send test email	brand_id="bnd123", template_name="ForgotPassword"	Test email sent to current user	Test delivery
TEMPLATE-POS-011	Get email settings	brand_id="bnd123", template_name="UserActivation"	Returns recipient settings	Recipient config
TEMPLATE-POS-012	Update email settings	brand_id="bnd123", template_name="UserActivation", recipients="ALL_USERS"	Settings updated	Recipient set changed
TEMPLATE-POS-013	Delete customization	customization_id="id123" with confirmation	Customization deleted	Single deletion
TEMPLATE-POS-014	Delete all customizations	brand_id="bnd123", template_name="ForgotPassword" with confirmation	All variants deleted	Bulk deletion

TEMPLATE-POS-015	Template with variable substitution	body contains "activationLink","activationLink","{firstName}"	Variables preserved	Required placeholders

3.1.6 Email Domains - Positive Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
EMAIL-DOMAIN-POS-001	List email domains	No parameters	Returns all email domains	All domains listed
EMAIL-DOMAIN-POS-002	List with brand expansion	expand_brands=True	Returns domains with brand objects	Embedded brand data
EMAIL-DOMAIN-POS-003	Get specific email domain	email_domain_id="edm123"	Returns single domain with DNS records	Valid domain ID
EMAIL-DOMAIN-POS-004	Create email domain	brand_id="bnd123", domain="example.com", user_name="noreply", display_name="Acme IT"	Domain created	New sender address
EMAIL-DOMAIN-POS-005	Domain with custom subdomain	validation_subdomain="mail" (default)	Domain uses default subdomain	Default routing
EMAIL-DOMAIN-POS-006	Domain validation records	email_domain_id="edm123"	Returns dnsValidationRecords	DNS config structure
EMAIL-DOMAIN-POS-007	Verify email domain	email_domain_id="edm123"	Verification initiated	Async verification process

EMAIL-DOMAIN-POS-008	Update email domain	email_domain_id="edm123", display_name="New Display"	Domain config updated	Metadata change
EMAIL-DOMAIN-POS-009	Delete email domain	email_domain_id="edm123" with confirmation	Domain deleted	Destructive operation

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3.2 Negative Scenarios

These tests verify that invalid or malicious inputs are rejected with appropriate error messages.

3.2.1 ID Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
ID-NEG-001	GET, PUT, DELETE	Invalid brand ID - path traversal	brand_id="../groups/00g123"	Rejected with InvalidOktaIdError	Security validation
ID-NEG-002	GET, PUT, DELETE	Invalid brand ID - forward slash	brand_id="bnd123/bnd456"	Rejected with InvalidOktaIdError	Path separator blocked
ID-NEG-003	GET, PUT, DELETE	Invalid brand ID - backslash	brand_id="bnd123\bnd456"	Rejected with InvalidOktaIdError	Windows path separator blocked
ID-NEG-004	GET, PUT, DELETE	Invalid brand ID - dot-dot	brand_id="bnd123..bnd456"	Rejected with InvalidOktaIdError	Traversal sequen

					ce blocked
ID-NEG-005	GET, PUT, DELETE	Invalid brand ID - URL encoded slash	brand_id="bnd123%2fbnd456"	Rejected with InvalidOktaIdError	URL encoding attack blocked
ID-NEG-006	GET, PUT, DELETE	Invalid brand ID - query string	brand_id="bnd123?filter=true"	Rejected with InvalidOktaIdError	Query delimiter blocked
ID-NEG-007	GET, PUT, DELETE	Invalid brand ID - fragment	brand_id="bnd123#admin"	Rejected with InvalidOktaIdError	Fragment delimiter blocked
ID-NEG-008	GET, PUT, DELETE	Invalid brand ID - empty	brand_id=""	Rejected with InvalidOktaIdError	Empty ID rejected
ID-NEG-009	GET, PUT, DELETE	Invalid brand ID - null	brand_id=null	Rejected with appropriate error	Null handling
ID-NEG-010	GET, PUT, DELETE	Invalid domain ID - path traversal	domain_id="../certificates/cert123"	Rejected with InvalidOktaIdError	Security validation

3.2.2 Parameter Type Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
PARAM-NEG-001	GET, PUT, DELETE	Brand ID not string	brand_id=12345	Rejected or coerced appropriately	Type checking
PARAM-NEG-002	LIST	Limit out of range - too low	limit=0	Clamped to minimum 1 or rejected	Boundary validation
PARAM-NEG-003	LIST	Limit out of range - too high	limit=500	Clamped to maximum 200 or rejected	Boundary validation
PARAM-NEG-004	LIST	Limit negative	limit=-10	Rejected or clamped to 1	Invalid range
PARAM-NEG-005	CREATE, PUT	Certificate source type invalid	certificate_source_type="INVALID_TYPE"	Rejected with specific error message	Enum validation
PARAM-NEG-006	PUT	Theme variant invalid	variant="INVALID_VARIANT"	Rejected with list of valid options	Enum validation
PARAM-NEG-007	PUT	Email recipient setting invalid	recipients="INVALID_SETTING"	Rejected with valid options	Enum validation

PARAM-NEG-008	CREATE, PUT	Language tag malformed	language="invalid-tag-format"	Rejected with IETF BCP 47 error	Format validation
PARAM-NEG-009	PUT	CSP mode invalid	csp_mode="invalid_mode"	Rejected with valid mode options	Enum validation

3.2.3 Naming and Uniqueness Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
UNIQUE-NEG-001	CREATE	Create brand with duplicate name	name="ExistingBrand" (already exists)	Rejected with "Brand already exists" error	Duplicate detection
UNIQUE-NEG-002	CREATE	Create custom domain duplicate	domain="login.example.com" (already exists)	Rejected with "Domain already exists" error	Duplicate check
UNIQUE-NEG-003	CREATE	Create email domain duplicate	domain="example.com" (already exists)	Rejected with "Email domain already exists"	Duplicate check
UNIQUE-NEG-004	CREATE, PUT	Create email customization	brand_id="bnd123", template_name="UserActivation", language="en" (already exists)	Rejected or updated (dependi	Language uniqueness

		duplicate language		ng on spec)	
UNIQUE-NEG-005	CREATE	Create template customization without brand_id check	No lookup performed	Warning logged if lookup skipped	Pre-creation validation

3.2.4 Format Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
FORMAT-NEG-001	CREATE, PUT	Invalid domain name	domain="not a domain"	Rejected with domain format error	Domain format validation
FORMAT-NEG-002	CREATE, PUT	Domain with reserved pattern	domain="drapp.subdomain.okta.com"	Rejected as reserved	Okta reserved domains
FORMAT-NEG-003	CREATE, PUT	Email format invalid	user_name contains special chars like "@"	Rejected or escaped properly	Email component validation
FORMAT-NEG-004	CREATE, PUT	Subject line empty	subject=""	Rejected - required field	Mandatory field validation
FORMAT-NEG-005	CREATE, PUT	Body content empty	body=""	Rejected - required field	Mandatory field validation

FORMAT-NEG-006	PUT	HTML page content empty	page_content=""	Rejected - required field	Mandatory field validation
FORMAT-NEG-007	PUT	Color format invalid	primaryColor="not-a-color"	Rejected with color format error	Color validation
FORMAT-NEG-008	PUT	Color format valid hex	primaryColor="#FF0000"	Accepted	Valid hex color
FORMAT-NEG-009	PUT	Color format valid RGB	primaryColor="rgb(255,0,0)"	Accepted or rejected (spec dependent)	RGB color format

3.2.5 Sensitive Information Handling - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
SECURITY-NEG-001	CREATE, PUT	Private key in error response	Attempt domain create with cert including private key	Private key NOT exposed in error message	Acceptance Criterion 2
SECURITY-NEG-002	LIST, GET, CREATE, PUT, DELETE	Secret token in logs	Any operation with auth token	Token NOT logged in debug output	No token exposure
SECURITY-NEG-003	GET, PUT	Certificate private portion	Get domain returns certificate	Certificate metadata returned,	Partial cert exposure

				private key omitted	
SECURITY-NEG-004	GET	User credentials in preview	Get email template preview	No credentials in preview output	Template safety

3.2.6 Status and Permission Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
STATUS-NEG-001	GET	Query failed domain status	domain_id with FAILED_TO_VERIFY status	Status returned as-is per Acceptance Criterion 3	Org status reflected
STATUS-NEG-002	GET	Query pending verification	domain_id with IN_PROGRESS status	Status returned as-is	Current state
STATUS-NEG-003	POST (verify)	Verified domain re-verification attempted	domain_id with VERIFIED status	Operations proceed or return appropriate message	Already verified handling
STATUS-NEG-004	POST (verify), GET	Domain already verified - DNS records	Get DNS records for verified domain	Message indicates "Domain is already verified" not redundant setup records	Acceptance Criterion 1
STATUS-NEG-005	LIST, GET, CREATE, PUT, DELETE	Insufficient permissions	Any tool called without proper permission	Okta API error returned as-is	Permission errors reflected

3.2.7 URL and Path Validation - Negative Scenarios

Test ID	CRUD Operations	Scenario	Input	Expected Outcome	Notes
URL-NEG-001	PUT	Invalid callback URL	sign_out_redirect_url="not a url"	Rejected with URL format error	URL validation
URL-NEG-002	PUT	Callback URL with unsafe protocol	sign_out_redirect_url="javascript:alert()"	Rejected - unsafe protocol	XSS prevention
URL-NEG-003	PUT	File path with traversal	file_path="../../sensitive_file"	Rejected with path traversal error	File access security
URL-NEG-004	PUT	File path absolute system path	file_path="/etc/passwd"	Rejected - absolute paths not allowed	System file protection

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3.3 Edge Cases

These tests verify behavior at boundary conditions and unusual but valid scenarios.

3.3.1 Boundary and Edge Case Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes

EDGE-001	Maximum brand list size	limit=200 (max allowed)	Returns up to 200 brands	Boundary limit
EDGE-002	Single character brand ID	brand_id="a"	Accepted if matches pattern, may fail on API call	Single char handling
EDGE-003	Very long brand ID	brand_id="a" * 500	ID validated, may fail on API call	Length constraints
EDGE-004	Unicode in brand name	name="Brand-北京-东京"	Handled appropriately	Unicode support
EDGE-005	Special characters in display name	display_name="O'Reilly & Associates"	Accepted and escaped properly	Special char handling
EDGE-006	HTML entities in subject line	subject="Hello & Welcome"	Entities preserved or encoded	Entity handling
EDGE-007	Email with plus addressing	user_name="noreply+test"	Accepted and handled correctly	Plus addressing support
EDGE-008	Domain with many subdomains	domain="auth.staging.api.example.com"	Accepted if valid format	Subdomain nesting
EDGE-009	Pagination with zero results	fetch_all=True on empty org	Returns empty list with pagination metadata	Empty result handling

EDGE-010	Cursor pointing to non-existent page	after="invalid_cursor_123"	Handled gracefully, error returned	Invalid cursor handling
EDGE-011	Very large HTML content	page_content with 10MB of HTML	Rejected with size limit error or truncated	Size constraints
EDGE-012	CSS with code execution	custom_css contains script tags	Properly escaped or rejected	XSS prevention
EDGE-013	Theme color edge values	primaryColor="#000000" or "#FFFFFF"	Accepted	Extreme colors
EDGE-014	Whitespace-only input	name=" "	Rejected or trimmed to empty	Whitespace handling
EDGE-015	Expand parameter with invalid value	expand=["invalid_expand_type"]	All brands listed because API ignores invalid expand type	Invalid expand validation

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3.4 Scale Scenarios

These tests verify behavior with large data sets and concurrent operations.

3.4.1 Scale Test Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
SCALE-001	Org with many brands	List brands with fetch_all=True when org has 500+ brands	All brands returned with pagination info	Pagination under load

SCALE -002	Pagination over many pages	fetch_all=True crossing 100+ pages	Tool respects page limit (200 pages default) and returns pagination_info with stopped_early flag	Pagination cap enforcement
SCALE -003	Brand with many customizations	List email customizations for template with 50+ language variants	All variants returned efficiently	Large result set
SCALE -004	Complex theme configuration	Theme with logo, favicon, background, multiple variants	Full config retrieved and updated	Complex object handling
SCALE -005	Long expansion list	expand=["themes", "domains", "emailDomain"] on all brands	All expanded data embedded	Multiple expansions
SCALE -006	Large HTML page content	page_content is 5MB of valid HTML	Accepted or rejected with appropriate size error	Size boundary
SCALE -007	Large image upload	Upload 2MB favicon file	Accepted (at limit)	File size boundary
SCALE -008	Many DNS records	Custom domain with 10+ DNS records	All records returned and presented clearly	Large record set
SCALE -009	Search with many matches	q="common" matches 200+ brands	fetch_all=True returns all matches with pagination	Search result scaling

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3.5 Error Recovery and Messaging

These tests verify that error messages meet acceptance criteria and enable user recovery.

3.5.1 Error Message Validation Scenarios

Test ID	Scenario	Input	Expected Outcome	Notes
ERROR-MSG-001	Brand not found	brand_id="bnd_nonexistent"	Error message: "Brand not found (brand_id: bnd_nonexistent)"	Clear identification
ERROR-MSG-002	Invalid brand ID	brand_id="../invalid"	Error: "Invalid brand_id: contains forbidden character or pattern '/'"	Specific issue identified
ERROR-MSG-003	Duplicate domain	domain="existing.com" already exists	Error: "A custom domain 'existing.com' already exists (id: dom123). Use list_custom_domains() to find it or choose a different domain."	Acceptance Criterion 1
ERROR-MSG-004	Duplicate brand name	name="ExistingBrand"	Error: "Brand name 'ExistingBrand' already exists. Use list_brands(q='ExistingBrand') to find it or choose a different name."	Actionable error message
ERROR-MSG-005	Already verified domain	domain_id="dom_verified" calling verify	Error: "Domain is already verified" not "Please add these DNS records"	Acceptance Criterion 1
ERROR-MSG-006	Invalid certificate source	certificate_source_type="INVALID"	Error: "certificate_source_type must be one of ['MANUAL', 'OKTA_MANAGED'], got: 'INVALID'"	Clear enumeration
ERROR-MSG-007	Missing required field	brand_id omitted from create	Error: "brand_id is required"	Clear requirement

ERROR-MSG-008	Path traversal attempt	brand_id="../../etc/passwd"	Error: "Invalid brand_id: contains forbidden character or pattern '..'"	Security incident handling
ERROR-MSG-009	Okta API error	API returns 429 (Too Many Requests)	Error: "Okta API error: Too many requests. Please retry after a delay."	Okta error passthrough
ERROR-MSG-010	Network connection error	Network timeout during API call	Error: "Failed to connect to Okta organization. Please check your connection and authentication credentials."	Troubleshooting help

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3.6 User Prompt Interpretation

These tests verify that the tools correctly interpret and handle the use-case prompts provided in the specification.

3.6.1 Use-Case Prompt Handling

Test ID	Scenario	Prompt	Expected Tool Call	Expected Behavior	Notes
PROMPT-001	Check brands using default templates	"Check all our brands. Which ones are currently using the default Okta email templates instead of our custom HTML versions?"	list_brands() + list_email_customizations() for each	Returns brands with template customization status	Query requires filtering logic
PROMPT-002	List failed domain verifications	"List all brands where the Custom Domain status is 'Failed' or 'Pending Verification'"	list_custom_domains()	Filters domains by status FAILED_TO_VERIFY or	Status filtering

				IN_PROGRESS	
PROMPT-003	Get theme config for reuse	"Get the full theme configuration (colors, background image, custom CSS) for the 'North-America' brand so I can use it to configure the new 'Asia-Pacific' brand."	list_brands() to find North-America, get_brand_theme(), get_brand()	Returns complete theme config for reuse	Template copying workflow
PROMPT-004	List active brands and find default	"List all the active brands in my Okta org and tell me which one is the default."	list_brands()	Returns list with isDefault flag identified	Default brand identification
PROMPT-005	Update brand theme color	"Update the 'Customer Portal' brand theme to use our new brand color"	list_brands() to find, replace_brand_theme()	Theme updated with new color	Theme update workflow
PROMPT-006	Show custom error page	"Show me the current HTML for our custom error page for the 'Partner' brand"	list_brands() to find Partner, get_customized_error_page()	Returns HTML content of error page	Page content retrieval
PROMPT-007	Reset sign-out pages	"Reset the sign-out page for all brands to use the default Okta custom template."	list_brands(fetch_all=True), delete_customized_sign_in_page() for each	Sign-out pages deleted/reset for all brands	Bulk reset operation
PROMPT-008	Check domain verification status	"What is the status of our custom domain auth.globalcorp.com? Is the TLS certificate verified?"	list_custom_domains(), filter by domain name, check validationStatus and certificate	Returns domain status and certificate state	Status verification

PROMPT-009	Switch domain for brand	"Switch our 'External Customers' brand to use the custom domain login.customers.com."	list_brands() to find, list_custom_domains() to find domain, replace_brand()	Brand's domain reference updated	Domain assignment
PROMPT-010	List template customizations	"List all the customizations we've made to the 'User Activation' email template."	list_email_customizations() for UserActivation template	All language variants listed	Customization enumeration
PROMPT-011	Update email subject	"Update the subject line of the 'Forgot Password' email to: 'Secure link to reset your [Brand Name] password'."	list_email_customizations() to find existing, replace_email_customization()	Subject updated with new text	Subject update
PROMPT-012	Create French template	"Create a French version of the 'Account Lockout' email template for our Canadian brand."	create_email_customization() with language="fr"	French customization created	New language variant
PROMPT-013	Check email domain verification	"Check if the email domain mail.marketing.com is fully verified and ready to send notifications."	list_email_domains(), find by domain name, check validationStatus	Returns VERIFIED or other status	Email domain status
PROMPT-014	Generate DNS records	"Generate the DNS TXT and CNAME records I need to add to my provider to verify our new custom domain."	create_custom_domain() returns dnsRecords	DNS records (TXT + CNAME) returned	Record generation

PROMPT-015	Setup email domain with SPF/DKIM	"Register no-reply@security.example.com as a new sender and give me the SPF and DKIM records to configure."	create_email_domain() returns DNS validation records	Email domain created with DNS records for SPF/DKIM	Sender setup

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4. Acceptance Criteria Validation Matrix

This section maps the specified acceptance criteria to test coverage.

4.1 Acceptance Criterion 1: Appropriate Error Messages

Criterion: "Error messages should be appropriate. For example, if the domain is already verified, the tool should return a status message stating 'Domain is already verified' rather than providing redundant setup records."

Test ID	Coverage	Validation
STATUS-NEG-004	Verified domain handling	Confirmed: Already verified domain returns "Domain is already verified" message
ERROR-MSG-003	Duplicate domain	Confirmed: Duplicate domain returns "already exists" message with actionable next steps
ERROR-MSG-005	Verified domain re-verification	Confirmed: Tool detects verified state and returns appropriate message
PROMPT-008	User prompt for verification	Confirmed: Status query returns current state without redundant records
ERROR-MSG-001 through ERROR-MSG-010	Error messaging framework	Confirmed: All error messages are clear, actionable, and specific

Test Coverage: 15+ tests specifically validate appropriate error messages.

4.2 Acceptance Criterion 2: Sensitive Information Protection

Criterion: "Sensitive information like secret keys/strings should not be passed to the LLM, even when queried."

Test ID	Coverage	Validation
SECURITY-NEG-001	Private key protection	Test: Domain creation with certificate does NOT expose private key in error responses
SECURITY-NEG-002	Token protection	Test: Authentication tokens NOT logged or returned in any response
SECURITY-NEG-003	Certificate handling	Test: Certificate metadata returned, private key components excluded
SECURITY-NEG-004	Template preview safety	Test: Email template previews do NOT contain embedded credentials

Test Coverage: 4 tests focused on sensitive data handling.

Implementation Notes:

- All error responses should be scrubbed of sensitive data before returning to LLM
- Certificate responses should only include public/metadata portions
- Logging should never include authentication tokens or API keys
- Template content returned to LLM should be sanitized of any embedded secrets

4.3 Acceptance Criterion 3: Organization Status Reflection

Criterion: "Obviously, the org status must be reflected as is in the queries according to user permissions."

Test ID	Coverage	Validation
STATUS-NEG-001	Failed domain status	Test: FAILED_TO_VERIFY status returned as-is from Okta
STATUS-NEG-002	Pending verification status	Test: IN_PROGRESS status returned as-is

STATUS-NEG-003	Verified status	Test: VERIFIED status returned as-is
STATUS-NEG-005	Permission errors	Test: Okta 403 Forbidden errors returned to user as-is
PROMPT-001 through PROMPT-015	Real status usage	Tests: All prompts work with actual org status values

Test Coverage: 5+ tests validate that organization status is never modified or misrepresented.

Implementation Notes:

- All status values (domain validation, brand status, template status) returned directly from Okta API
- Permission-based errors from Okta API should be passed through unchanged
- No client-side filtering or modification of permission-based responses

4.4 Acceptance Criterion 4: No Redundant Resource Creation

Criterion: "Redundant brand, templates names should not be created, a lookup to the existing set should be done to verify names before creation."

Test ID	Coverage	Validation
UNIQUE-NEG-001	Brand duplicate check	Test: Existing brand lookup before create_brand()
UNIQUE-NEG-002	Domain duplicate check	Test: Existing domain list queried before create_custom_domain()
UNIQUE-NEG-003	Email domain duplicate check	Test: Existing email domain list queried before create_email_domain()
UNIQUE-NEG-004	Template language duplicate	Test: Existing customizations checked before create_email_customization()
UNIQUE-NEG-005	Pre-creation validation	Test: Logs warning if lookup is skipped

Test Coverage: 5 tests validate duplicate prevention.

Implementation Details:

- `create_brand()` must call `list_brands()` or equivalent before creation to check for duplicates
- `create_custom_domain()` must call `list_custom_domains()` before creation
- `create_email_domain()` must call `list_email_domains()` before creation
- `create_email_customization()` must query existing customizations to prevent language duplicates
- Error message must clearly indicate existing resource ID and guide user to use it instead

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5. Test Execution Plan

Test execution follows these phases:

Phase 1: Positive Scenario Validation

- **Focus:** Verify valid prompts produce expected tool behavior
- **Method:** Manual review of test cases against tool responses
- **Coverage:** All positive scenarios (BRAND-POS through EMAIL-DOMAIN-POS)
- **Deliverable:** Validation checklist with pass/fail results

Phase 2: Negative Scenario Validation

- **Focus:** Verify invalid prompts are rejected with appropriate errors
- **Method:** Manual testing of malicious/invalid inputs
- **Coverage:** All negative scenarios with CRUD operation specificity (ID-NEG through URL-NEG)
- **Deliverable:** Documentation of error messages and responses

Phase 3: Edge Case Validation

- **Focus:** Verify boundary conditions are handled correctly
- **Method:** Manual testing of extreme but valid inputs
- **Coverage:** All edge case scenarios (EDGE-001 through EDGE-015)
- **Deliverable:** Results showing how tool handles boundaries

Phase 4: Prompt Interpretation Validation

- **Focus:** Verify user prompts are interpreted correctly
- **Method:** Manual validation of the 15 use-case prompts
- **Coverage:** PROMPT-001 through PROMPT-015 scenarios
- **Deliverable:** Prompt interpretation test matrix with results

6. Risk Assessment and Mitigation

6.1 Identified Risks

Risk	Impact	Probability	Mitigation
Duplicate creation slip through	Data integrity issue	Medium	Implement pre-creation lookup tests (UNIQUE tests)
Sensitive data exposure	Security breach	High	Dedicated security tests (SECURITY-NEG tests)
Path traversal attack	System compromise	High	ID validation tests (ID-NEG tests)
Incorrect error messages	User confusion	Medium	Error message validation tests (ERROR-MSG tests)
Pagination breaks on large orgs	Incomplete data	Medium	Scale tests (SCALE tests)
Status misrepresentation	Policy violation	Low	Acceptance Criterion 3 tests (STATUS tests)

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8. Test Coverage Summary

8.1 Coverage by Tool

Tool	Positive	Negative	Edge	Scale	Error Msg	Total
Brands	11	8	2	1	2	24
Themes	13	8	3	1	2	27
Custom Pages	18	8	3	1	2	32

Custom Domains	10	10	3	2	2	27
Email Templates	15	8	3	1	2	29
Email Domains	9	8	2	1	2	22
Cross-Tool	-	-	3	3	-	6

Total Test Cases: 167+ (excluding duplicate variations)

8.2 Coverage by Category

Category	Test Count	Priority	Status
Input Validation	35	Critical	Phase 1
Duplicate Detection	5	Critical	Phase 1
Error Messages	10	High	Phase 2
Security	4	Critical	Phase 3
Prompt Handling	15	High	Phase 4
Edge Cases	15	Medium	Phase 1-2
Scale Scenarios	9	Medium	Phase 2
Status Handling	5	High	Phase 2